Fusing WiFi Signals and Camera for Driver Activity Recognition based on Deep Learning

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***Abstract*—In-vehicle camera has been widely adopted for driver activity recognition. Unfortunately, camera-based solutions have some limitations, like brightness condition, blurriness caused by bumpy roads, and angle of view. To this end, some recent studies have resorted to other emerging recognition technologies that show great promise, such as the Channel State Information (CSI) of WiFi signals. However, it is well-known that WiFi signals can be interfered significantly by other nearby WiFi systems, such as neighboring vehicles on the same road. Hence, it is important to address this challenge for WiFi-based solutions to become more practical.**

In this paper, we propose DriveFusion that overcomes the limitations of both WiFi-based solutions and in-vehicle camera by fusing their inputs based on deep learning. Specifically, DriveFusion flexibly extracts useful information from both systems by having 1) a novel non-reference CSI signal quality estimator based on network analysis, 2) a learning-based signal filter to exclude low-quality features, and 3) an online feature combination module that assigns weights to the two inputs dynamically, based on their signal qualities that can change rapidly. Our results in a real car with commercial WiFi devices show that DriveFusion achieves a recognition accuracy of 91.3% when the deployed WiFi interference is relatively strong, which outperforms state-of-the-art baselines by 13.4% on average.

I. Introduction

Recognizing driver activities is an important task for enhancing road safety or driver assistance. For example, a distracted driver can get helped by receiving a warning sound upon the detection of turning her/his head to the back for too long. Likewise, the gestures of a disabled driver can be utilized to help control the car entertainment system. Existing work on driver activity recognition focuses mostly on camera because it is cost-efficient and easy to be installed in a vehicle [1][2][3]. However, in addition to privacy concerns, a camera system has several key limitations, such as blurriness caused by bumpy roads and different brightness conditions, which make the recognition performance sensitive to road and weather conditions. Although infra-red and thermal cameras have partially overcome the limitations of the darkness and illumination at increased costs, camera still has problems such as obscurity caused by direct sunlight or driver motions, and edge erosion [4][5]. Moreover, the current image processing algorithms still face some challenges such as partial occlusion and object overlapping [6][7]. Thus, using camera alone in a vehicle may not lead to the best performance for driver activity recognition.

On the other side, emerging wireless recognition technologies, such as WiFi sensing and acoustic sensing, have shown great promise in driver activity recognition. These recognition schemes can overcome the limitations of traditional camera-based methodologies [4][8][9][10], including brightness condition, blurriness, and angle of view. In addition, compared with solutions using Inertial Measurement Unit (IMU) sensors, these recognition systems are also non-intrusive and do not require users to buy and wear any device on his/her body. However, WiFi or acoustic-based activity recognition also has its own limitations. For example, WiFi sensing can suffer from the contention of wireless channels. Since WiFi signal is sampled on a packet basis, WiFi-based solutions often require that packets be sent at a speed of 400 packets/s or higher to yield an accurate recognition result [11][12]. Although the interference of other WiFi systems can be mitigated by the metal enclosure of a vehicle, it cannot be neglected when there happens to exist other WiFi transmitters on the same channel in the close surroundings. In such situations, the severe contention for the shared wireless channel can make a WiFi-based recognition system fail to maintain such a high packet sending frequency, due to the need of packet retransmissions. As a result, the recognition accuracy can be degraded significantly. Similarly, acoustic sensing can also be significantly interfered by the noise from nearby acoustic devices [13]. Since WiFi and acoustic signals have similar problems, in the following, we focus on WiFi sensing as an example to simplify the discussion, but our solution can be applied to acoustic sensing as well.

To overcome the limitations of both camera-based and WiFi-based solutions, one methodology is to combine the signals from the camera and WiFi together as an integrated activity recognition system. One may think such an integration is trivial, because we could simply switch to WiFi-based system whenever it is dark, and switch to camera whenever WiFi encounters high contention. Unfortunately, such a simplistic solution would *not* work well for three reasons. First, camera can have poor recognition results due to causes other than darkness, such as direct sunlight, motion obscurity, and edge erosion. On the WiFi side, interference-free signals are lacking as references. Hence, it is difficult to estimate the quality of WiFi/camera input and judge when to switch to which system. Second, due to the mobility of vehicles, those conditions often change fast over time. For example, a car can approach a roadside WiFi transmitter and depart quickly. Hence, dynamic and fast adaptation is required for the integration solution. Last and most importantly, relying on only one of the two systems at a time is a waste of useful information provided by the other one. For example, though we have severe wireless contention and so the WiFi-based system might yield partially poor result,

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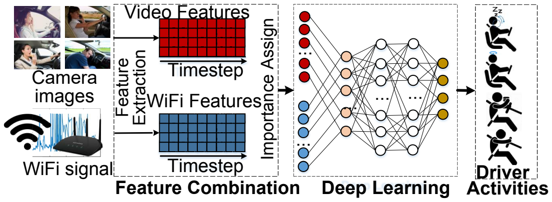


Fig. 1: The concept of DriveFusion. DriveFusion combines the camera input and WiFi signal with deep learning techniques.

some of its good results can still be utilized for improving the overall recognition accuracy.

To address these challenges, we propose DriveFusion, a WiFi-camera combined driver activity recognition system based on deep learning. First, in order to estimate the WiFi CSI quality, an online signal quality estimation algorithm with regression models is designed based on network congestion and delay analysis, because we cannot directly use well-known metrics such as Signal-to-Interference-plus-Noise Ratio (SINR) to estimate the signal quality without a reference. Second, unlike state-of-the-art solutions that are designed mainly to leverage stable inputs, DriveFusion aims at the scenarios where the quality of the input signals can be time-varying (e.g., due to vehicle mobility). In order to filter out misleading information caused by fast-varying signal qualities, we propose a dynamic and learning-based threshold determination method to exclude low-quality features from both inputs. Finally, to extract useful information from both WiFi and camera, DriveFusion includes a feature combination algorithm, designed based on the self-attention mechanism [14], to fuse the heterogeneous inputs from WiFi and camera with online importance assigning. After that, DriveFusion is designed based on Recurrent Neural Network (RNN) with the Long Short-Term Memory (LSTM) model [15] to flexibly process the combined inputs from both WiFi devices and camera and output the recognition result. Also, DriveFusion dynamically adjusts the importance assigned to the two inputs for handling fast-varying interference conditions.

The concept of DriveFusion is shown in Figure 1. The raw signal from WiFi devices and input from the camera are first denoised. Then, the characteristics representing the ongoing activity are extracted as the feature vectors from these denoised inputs. After that, deep learning is applied to the feature vectors, combines these vectors based on the assigned importance, outputs the sequence of the combined features for the ongoing activity, and finally recognizes the activity. After the activity is recognized, DriveFusion can send warnings to the control system of the vehicle if the recognized activity is related to dangerous driving behaviors. Specifically, the contributions of our paper are listed as follows:

- While existing work relies on either camera or WiFi for driver activity recognition, we propose DriveFusion that fuses the CSI signal and camera image features based on deep learning. To our best knowledge, our work is the first one that combines camera and WiFi for improving the accuracy of activity recognition.
- The design of DriveFusion features three novel algorithm designs: 1) a novel non-reference CSI signal quality estimator based on network analysis, 2) dynamic and learning-based threshold determination method to filter out low-quality features, and 3) an online feature combination module that dynamically assigns importance to handle fast-varying inference conditions.
- We evaluate DriveFusion with more than 800+ driver activity traces with both image sequence inputs and CSI signal traces with different image qualities and wireless interference levels. The evaluation demonstrates that DriveFusion can achieve a recognition accuracy of 91.3%, and outperform state-of-the-art data fusion schemes by 13.4% when relatively strong interference is deployed.

The rest of this paper is organized as follows. Section II discusses the related work. Section III motivates DriveFusion by testing the existing WiFi-based driver activity recognition system under wireless interference. Section IV presents the design of DriveFusion. Section V shows the evaluation results of DriveFusion. Section VI concludes the paper.

II. Related Work

With the advance of the autonomous driving and intelligent traffic, driver activity recognition has become an important application. Among these recognition systems, a traditional methodology is camera [1][16][17]. Recently, there are some studies on in-car driver activity recognition using the CSI signal with commercial WiFi devices [4][8][10][9][18]. For example, WiFind uses the CSI signal amplitude and phase to detect whether the driver is fatigued or not based on breathing rate monitoring and activity recognition [8]; WiDrive proposes real-time driver activity recognition, enabling an activity to be recognized even before it finishes [10]. ViHOT monitors the head rotation angle of the driver using the CSI phase information based on the Dynamic Time Warping (DTW) distance calculation [9]. However, all the aforementioned solutions leverage either camera or CSI, but not both. As a result, they may result in a poor recognition accuracy in some conditions, such as insufficiently recorded CSI samples due to wireless interference or blurred camera images.

There are some studies that try to fuse the inputs from heterogeneous sensors [19]. Some early work of data fusion uses simple strategies such as feature concatenation [20][21]. However, such simple methods often cause over-fitting in the case of a small size training sample, and are difficult to discover non-linear relationships in different input signals [19]. Recently, some studies begin to use deep learning for cross-domain sensor fusion, which can automatically determine how to extract features from different sensors and combine them without manual reasoning and parameter tuning, such as DeepSense [22], DeepFusion [23], and GlobalFusion [24]. However, those studies are designed for wearable or wireless sensors and for in-room scenarios, where the recognition performance of the involved sensors is relatively stable. As a result, these solutions cannot be directly applied to driver activity recognition where the wireless contention and image quality can change fast and dramatically due to the vehicle mobility. In sharp contrast, our work is proposed to integrate the inputs from WiFi and camera for an in-car scenario with a large signal quality variation, and explicitly considers the signal qualities in its data fusion procedure.

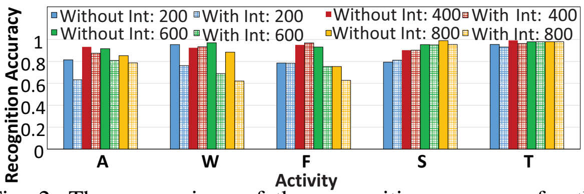


Fig. 2: The comparison of the recognition accuracy for the tested CSI-based solution with and without wireless interference (Int) from other vehicles (e.g., *With Int: 200* means with interference at 200Hz). The average accuracy drops by 7.2% when the interference is switched on.

III. Motivation

To test the impact of the wireless interference on WiFi-based driver activity recognition, we select and implement a state-of-the-art solution called WiDrive [10] in our motivation study. WiDrive is also used as a baseline later in our evaluation. We conduct an experiment with two vehicles in a parking lot. Both of them are equipped with a pair of wireless transmitter and receiver. The packet sending frequency for each transmitter is set to be 800Hz and can be tuned to different values during the experiment. Since the CSI value is recorded whenever a packet is received, we define the CSI sampling frequency as the number of received packets per second on the receiver side. Note that the actual CSI sampling frequency on the receiver side can be different from the packet sending frequency used on the transmitter side due to wireless channel contention. In order to emulate different levels of channel contentions with only two working transmitters, we deliberately enlarge the sending packet length of the transmitter to the receiver to 1024 bytes on both vehicles. The CSI values are recorded in one car when the other car serves only as the interference node. Here we test the recognition accuracy of five typical driver activities that are commonly used in related work [10], [4], [18]: Approaching the driving wheel (**A**), withdraw from the wheel (**W**), fetching items (**F**), swipe (**S**), and head turning (**T**). We first test the recognition accuracy when the interference is on, and then switch off the transmitter in the interfering vehicle to test the recognition accuracy without interference for comparison.

Figure 2 shows that compared with the scenario without wireless interference, the average recognition accuracy of the tested WiFi-based driver activity recognition system drops by 7.2% when the wireless interference is on. Interestingly, if both the two vehicles use the default packet sending frequency of 800Hz, the recognition accuracy is not good when there is interference. For example, at 800Hz, the recognition accuracy is 5.2% lower than the case when the packet sending frequencies of the WiFi transmitters in both vehicles are set to 400Hz. This is because the contention for the wireless channel between the two vehicles becomes more severe with a higher sending frequency, leading to a longer waiting time in the Carrier-Sense Multiple Access (CSMA) backoff mechanism. As a result, CSI packets cannot be received at the sending frequency (i.e., 800Hz) in the Media Access Control (MAC) layer due to the interference, which leads to a smaller than desired CSI sampling frequency. Consequently, during recognition, more data points have to be estimated based on linear interpolation between two consecutive CSI records [12]. These estimated data points cause potential information loss on the ongoing activity, thus a lower recognition accuracy.

On the other hand, unfortunately, simply lowering the CSI sending frequency on the test vehicles cannot solve this problem either: As shown in Figure 2, if the sending frequency is too low (e.g., 200Hz), the accuracy also drops because a too low CSI sampling frequency cannot capture the Doppler shift caused by the human movement due to the aliasing effect, so a portion of the frequency shift caused by the activity is lost. Hence, simply reducing the packet sending frequency cannot solve this problem and may cause performance degradation due to the lack of CSI samples.

In summary, the recognition performance of the WiFi-based driver activity recognition system indeed degrades due to the wireless channel contention. This degradation cannot be solved by simply increasing or decreasing the packet sending frequency. Moreover, with more vehicles on the future road, this problem can become more severe because the CSI sampling frequency becomes even lower due to more intense contention, and more vehicles would be impacted. During rush hours when the traffic density can be 140 vehicles per mile per lane (which corresponds to 43 vehicles in the communication range of WiFi transmitters per lane) [25], the recognition result provided by a WiFi-based system can be unreliable for all the vehicles due to the severe channel contention. This experiment provides a strong motivation that a WiFi-base recognition system cannot work well alone due to the strong wireless interference on the road. Thus, we must increase the robustness of such a system by integrating with another recognition methodology (e.g., camera).

IV. Design of DriveFusion

To address the interference problem, we propose to fuse the CSI signal with camera for an integrated driver activity recognition system, because in-vehicle camera is already widely adopted as part of the Advanced Driver Assistance System (ADAS). In the following, we first introduce the overview of DriveFusion and then each component in detail.

A. Design Overview

DriveFusion consists of both an offline training phase and an online processing phase. The offline training phase of DriveFusion first processes the training data as the input to a deep learning network. Specifically, the CSI signal and camera input shall be processed separately: The camera input is converted to a series of images on every frame, and then an individual Convolution Neural Network (CNN) is applied to each captured image to extract the image feature; The CSI signal is first denoised using conjugate multiplication proposed in WiDance [26]. Then, Principle Component Analysis (PCA) is applied to reduce the dimension of CSI signals. The processed CSI signal is transformed into spectrogram using time/frequency analysis, and then segmented to the feature vector in each time window. The length of the time window is the same as the frame duration of the image sequence input, so the numbers of the CSI feature vectors and the image feature vectors are equal (referred to as timestep in the rest of the paper) for an activity, in order to guarantee that there are

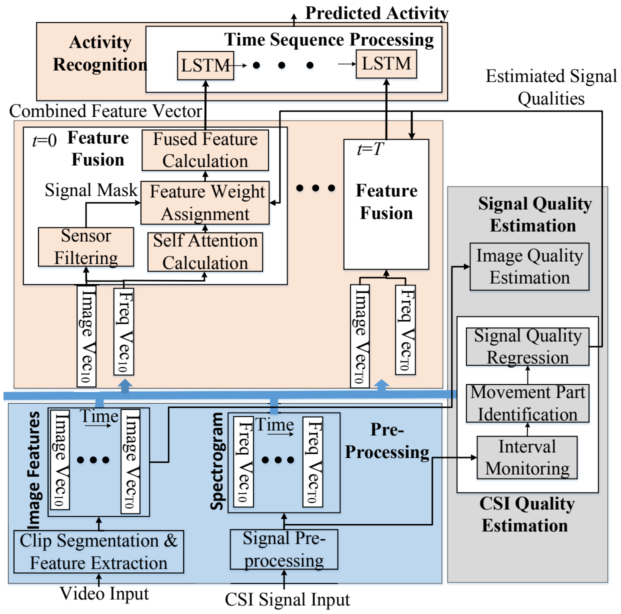


Fig. 3: Framework of DriveFusion. DriveFusion consists of four main components: data pre-processing, feature fusion, signal quality estimation, and activity recognition.

both camera input feature and CSI feature to be combined on each timestep for the ongoing activity. After that, the deep learning framework is trained using the processed input data of both CSI signal and camera images with different signal qualities. To estimate the signal qualities, we also build two separate regression models dealing with the CSI signal and camera input offline, respectively. The regression model of CSI signal is trained based on network analysis; the regression model of the image quality estimation is trained based upon the existing studies on patch-based image quality assessment algorithm [27].

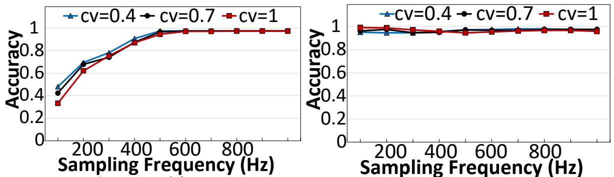
The online processing phase of DriveFusion is shown in Figure 3. Generally, there are four major steps for WiFi-camera based driver activity recognition: 1) The CSI signal and camera input shall be processed separately in the same way as in the aforementioned offline data pre-processing phase. 2) The processed CSI signal and the extracted image features serve as inputs to *signal quality estimation*, where the qualities of the camera input and CSI signal are estimated based on the offline-trained regression model for each input. 3) *Feature fusion* then takes the feature vectors of the CSI signal, the image feature on each timestep, and the estimated signal qualities as the inputs and begins the feature combination process. Feature fusion shall first decide whether to use each feature from either input signal for feature combination, and filter out those features whose signal quality are too low, based on a threshold learned in signal filtering. Then the feature fusion component calculates the signal importance based on the self-attention mechanism and the estimated signal qualities, and then combines the feature vector using the weighted average of all the remaining features. 4) The *activity recognition* component then leverages the Long Short-Term Memory (LSTM) to process the sequence of combined features. The activity can then be recognized with several fully-connected layers based on the likelihood with each pre-defined activity type.

The technical novelty of DriveFusion mainly lies in three aspects: 1) A regression-based method to estimate the quality of the CSI signal online, 2) a dynamic and learning-based threshold determination scheme for signal filtering, and 3) a dynamic weight assignment algorithm used in the fusion of CSI and images to handle fast-changing signal quality. In the following, we introduce them in more detail.

B. Signal Quality Estimation

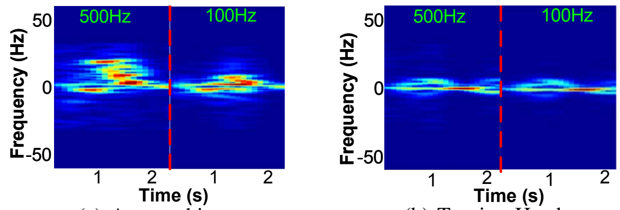
Signal quality estimation plays an important role in DriveFusion, where feature fusion leverages its output to filter out these signal inputs that cannot improve the overall recognition performance, and sets the weights for the input features using the self-attention mechanism. As a camera-WiFi combination system, quality estimation considers both the camera and CSI input. For the camera signal estimation, DriveFusion adopts the idea from the image quality assessment studies such as the patch based algorithm [27]: The image feature is first extracted by a CNN, and a regression model is then trained using this feature and outputs the quality with a range of zero (totally unrecognizable) to one (without any quality degradation). On the other side, the CSI signal quality is more difficult to estimate: First, there is no reference signal available that is free from the wireless interference. As a result, some standard indicators such as Signal-to-Interference-plus-Noise Ratio (SINR) cannot be calculated directly. Even though we can have the ideal CSI signal offline with no interference, the online collected CSI signal is still different than the offline ones due to different activity scales and hardware settings. Second, unlike image signal estimation, it is hard to train a similar regression model based on the CSI input features (i.e., the spectrogram), because the training process of the regression model requires manual labeling of the trained data: Unlike images whose quality can be estimated and related to the activities easily, the Doppler shift of the CSI spectrogram is much difficult for people to relate it to a certain activity and estimate its quality by their visual perception.

Since the CSI signal is sampled on a packet basis, we investigate how wireless interference impacts the CSI signal from network analysis perspective to perform CSI signal quality estimation. From the analysis of the 802.11 based network protocol and the Distributed Coordination Function (DCF) backoff mechanism, the network performances that are most related to activity recognition are the packet transmitting rate and the packet error rate (as shown in the motivation test). With the increase of other WiFi transmitters, the packet shall be waiting for a longer time in the transmitting queue, making the CSI signal be sampled at a lower frequency. As a result, some fast changing Doppler frequency shifts in the CSI signal cannot be captured in the spectrogram; A higher packet error rate makes more packets be re-transmitted, which increases the variance to the CSI sample frequency and introduces background noise in the spectrogram. Thus, we can treat the recognition accuracy when using the CSI signal alone as the signal quality, and investigate the relationship between the recognition accuracy and those two factors (i.e., packet sending rate and error rate) to estimate CSI signal quality.



Approaching Turning head

Fig. 4: The recognition accuracy of using CSI and its coefficient of variance (*cv*) under different sampling frequencies.



(a) Approaching (b) Turning Head

Fig. 5: The spectrogram of approaching driving wheel and turning head under two different sampling frequencies.

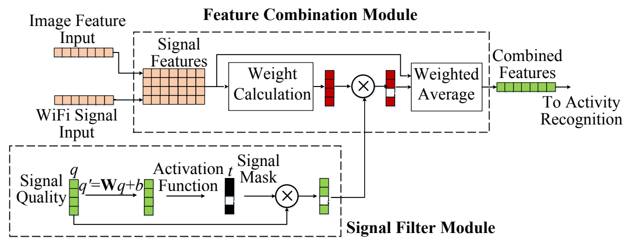


Fig. 6: The structure of feature fusion on each timestep. The major modules are feature combination and signal filtering.

On the receiver side, the CSI sampling interval (reciprocal of the sampling frequency) and its variance can be directly measured. Thus, we can relate the average CSI sampling interval and its variance to the overall recognition accuracy when applying only CSI signal to do the recognition, and build a regression model based on these two factors instead of packet sending rate and error rate. To build the regression model, we collect a set of five typical driver activities in the same way as in the motivation section, and deliberately down-sample the CSI signal and add variance. We normalize the variance to the coefficient of variance (*cv*), which is calculated as the standard variance divided by the mean of the CSI sampling interval. In such a way, the normalized variance with different means of the CSI sampling interval can be comparable to each other. Two typical activity results (approaching the driving wheel and turning head) are shown in in Figures 4(a) and (b).

From Figure 4, we can see that different activities have different sensitivities to the interference. For the approaching activity, when the sampling frequency becomes lower than 400Hz, the decrease speed of recognition accuracy becomes larger when we further decrease the CSI sampling frequency. This phenomenon is mainly caused by the loss of the low-frequency Doppler profile due to a too low CSI sampling frequency. Because most in-car activities are small-scaled, if these low-frequency features are lost due to a too low CSI sampling frequency, it shall have a much larger impact than the case where only relative high Doppler shifts are lost. As a result, the approaching activity is more impacted by the decrease of the CSI sampling frequency: an 82.1% drop with a CSI sampling frequency of 50Hz compared with a 500Hz sampling frequency. On the other hand, the recognition accuracy of the turning head activity does not show a large decrease, with only an 1.4% decrease. The reason for this is that the movement scale of arm movements is usually larger than their head counter parts, requiring a higher sampling frequency to recognize them correctly. Figure 5 shows the spectrograms of an approaching and turning head activity at sampling frequencies of 500Hz and 100Hz. We can see that with the decrease of the CSI sampling frequency, some Doppler shift of the approaching activity becomes much less apparent in Figures 5(a), while the spectrograms of turning head activity in Figures 5(b) are not impacted significantly.

To address the issue that different activities have different levels of sensitivity to wireless interference, we build two CSI quality estimation models separately depending on the moving part of the human body: the arm and the head, because the arm movement usually incurs a larger Doppler frequency shift than the head movement. To differentiate the arm and head movements, we use two separate cameras to record the head and arm movement separately. When the driver moves his/her arm or head, the sudden changes in the captured image can be known by either camera (if both cameras capture the movement, the ongoing activity is treated as the arm movement because the arm movement has a larger moving scale). The camera can then notify the CSI quality estimation component, and the corresponding quality estimation model can be loaded to do the signal estimation. We also integrate existing methods such as the one proposed in WiBot to distinguish head and arm movement with only WiFi signal [4] when camera images are not reliable, or use them both to increase the robustness of this coarse-grained activity recognition.

C. Feature Fusion

Feature fusion has two major modules: signal filtering and feature combination. As shown in Figure 6, signal filtering takes the inputs from signal estimation and filters out these input signals whose estimated signal qualities are too low to be combined; Feature combination takes the features from the camera input and the CSI signal, and calculates the importance of each signal that is not filtered by signal filtering based on the estimated signal quality and the attention model, in order to combine the signals in a moving environment.

Signal Filtering. Signal filtering aims to filter out those signal inputs that cannot improve the overall recognition performance in feature combination. It takes the estimated signal quality vectors of both CSI and camera as the input (q in Figure 6), and performs a linear transformation on the signal quality vectors (q' in Figure 6). After that, a sigmoid activation function is applied to the transformed signal quality vector, and outputs a binary mask for the input signals (t in Figure 6). The binary mask is then multiplied by the original signal quality to compute the filtered signal qualities. Specifically, signal filtering is built based on leveraging the estimated signal quality, which can be seen as a direct indicator of whether an input signal can be used in the recognition task or not: If the quality of one input signal is too low, it is highly possible that the extracted features of this signal cannot provide useful information in the fusion procedure. Thus, we choose a threshold-based mechanism in signal filtering: It aims to find all the input features whose signal qualities are above a certain threshold, and only lets the extracted features of these input signals be combined in feature combination.

However, how to set up the threshold is non-trivial. First, the relationship between useful information in the feature vectors and the signal quality is implicit and cannot be known directly. It is difficult to determine a good threshold for signal quality, such that any input signal with an estimated quality below it is guaranteed to contain only misleading information and should be discarded. Second, thresholds for different input signals are often dependent on each other and should be adjusted online based on the current recognition performance. For example, if the WiFi-based recognition system already shows a high recognition accuracy, it also requires the camera-based system to have a relatively good performance to further increase the overall performance. On the contrary, if the performance of the WiFi-based recognition system is poor, the requirement of the image quality is not as high as the previous case, because there is large space for performance improvement and combining the low-quality camera inputs can still increase the recognition performance. Thus, setting fixed thresholds manually for the input signals offline is not an appropriate option.

To address this challenge, we propose to determine the thresholds based on the training data and a deep learning structure. As shown in Figure 6, signal filtering is an essential part of DriveFusion with trainable parameters related to the thresholds values. If we can relate the quality thresholds with these trainable parameters, the thresholds used in the signal filtering can be determined automatically by the training data afterwards: To optimize the overall loss function, the optimizer shall determine those trainable parameters with the forward-backward algorithm during the training process of the whole deep learning network structure [28]. Mathematically, the settings of signal filtering and its trained parameters can be stated as follows: Given the input signal estimator $q = [q_1, q_2, ..., q_n]$, we first apply a linear transformation to q as follows:

$$q' = \mathbf{W}q + b \tag{1}$$

Equation (1) can be seen as a shifting operation to the original quality estimator q, where matrix $\mathbf{W}$ and vector b are the parameters to be trained. Matrix $\mathbf{W}$ can capture the dependence of the signal qualities on each other among different input signals. As a result, the threshold for one input signal does not only depend on its own value, but is also impacted by the qualities of other input signals due to the non-zero, non-diagonal elements in matrix $\mathbf{W}$. The learned thresholds are implemented by applying the step activation function to q', which ensures that the signals with $q'_i < 0$ are not activated and assigned with zero values (i.e., filtered out). **Example**: Suppose DriveFusion has one input CSI signal and one camera, and the estimated quality of the CSI signal is q_{csi} and the image quality is q_{cam}. Thus, in Equation 1 we have $q = \begin{bmatrix} q_{csi} \\ q_{cam} \end{bmatrix}$; If the learned parameters by the optimizer are $W = \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, the CSI signal is not filtered out if and only if:

$$W_{11}q_{csi} + W_{12}q_{cam} + b_1 > 0 \tag{2}$$

and the same logic applies to the camera input as well:

$$W_{21}q_{csi} + W_{22}q_{cam} + b_2 > 0 \tag{3}$$

Specifically, we use the sigmoid activation function instead of the step activation function because the ideal step activation is non-differentiable due to the discontinuity issue when $q' = 0$, and cannot be processed in the deep learning framework. Thus, the mask t_i for the i^{th} signal is calculated as: $t_i = \frac{1}{1+e^{q'_i}}$, where t_i indicates whether the corresponding feature vector of the i^{th} signal can be used in feature combination or not: If t_i for the i^{th} signal is non-zero, it indicates that the corresponding feature vector is not filtered out, and its signal quality shall be used directly in feature combination; Otherwise, if t_i of an input signal is zero due to a low quality estimator q_i, the corresponding feature shall not be combined, and have no influence on the activity recognition.

Feature Combination. After filtering out these features with signal qualities below the learned thresholds, feature combination is applied to remaining features of the input signals. Different from existing work, feature combination in DriveFusion not only assigns the importance of each input feature by the neural network model offline, but also considers the online signal quality estimation. Specifically, given the input CSI and image features and the filtered signal qualities, feature combination performs three major steps: 1) calculates primal weight for each signal feature with the self-attention mechanism, 2) multiplies the weights by the output from signal filtering, and 3) combines the features by weighted average.

Given the input feature vectors, feature combination first calculates the importance of each signal based on the attention mechanism, which is actively researched in the field such as computer vision and disease prediction [29][30]. The attention mechanism is inspired by human conceptual systems: human beings always pay attention to a certain region of an image during recognition, and adjust the focus over time. In our scenario where camera and WiFi signals can be seen as two separate regions of a whole image, and the importance assigned to them should also be changing over the whole activity duration. Thus, the attention mechanism can be applied to the input feature vectors to determine whether we should focus more on the image feature input, or the CSI Doppler shift input based on the same logic. We choose the self-attention mechanism to calculate the weight for each input signal in our scenario. The self-attention mechanism does not require any prior knowledge of the input activity, and has been shown to be promising in the machine learning research such as machine translation and activity prediction [14][15]. Mathematically, the procedure of calculating the weight of each input signal using the self-attention can be stated as follows: Given the feature vectors $F_1, F_2, ..., F_i$ of the signals from 1 to i, the weight α_i assigned to each feature input by the self-attention mechanism can be formulated as:

$$\begin{aligned} u_i &= tanh(\mathbf{W}_a F_i + b_a) \\ \alpha_i &= \frac{e^{{u_i}^T w_c}}{\sum_i e^{{u_i}^T w_c}} \end{aligned} \tag{4}$$

where matrix $\mathbf{W}_a$, vector b_a, and vector w_c are the parameters to be trained. The main idea of Equation (4) can be seen as follows: we first calculate the hidden representation of F_i through a one-layer perceptron to get u_i. We then measure the weight of the i^{th} sensor as the similarity of u_i with a sensor-level context vector w_c, and get a normalized weight α_i through a softmax function. Compared with a baseline (called Alternate and defined in Section V-A) that alternatively uses either CSI or camera, DriveFusion outperforms this baseline because the importance assigned by DriveFusion can be any fractional value between zero and one with the self-attention mechanism, whereas Alternate can only assign either zero or one.

Normally, the combined feature vector F_c can then be calculated as the weighted average according to α_i, i.e., $F_c = \sum_i \alpha_i F_i$. However, the scheme above does not consider the signal qualities at runtime. To incorporate the signal quality into DriveFusion, we further moderate the weight α_i assigned to the feature of the i^{th} signal as:

$$\alpha_i' = q_i t_i \alpha_i \tag{5}$$

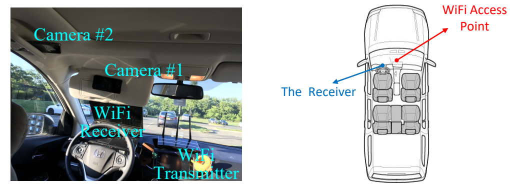
where t_i is the mask calculated in signal filtering for the i^{th} input signal, q_i is the input signal quality from signal estimation. After calculating the weights α_i to the CSI and image features, the combined feature F_c is calculated as a weighted average, i.e., $F_c = \sum_i \alpha_i' F_i$. With the above moderation, DriveFusion not only considers the parameters learned within the feature vector itself, but also takes the CSI quality as part of weight calculation. In such a way, even if a feature vector has a small weight from the self-attention mechanism, it can still play an important role in the combination as long as other signals are noisy and not reliable.

D. Discussion

Impact of the metal car body on the CSI interference. One may think that the metal car body can block wireless signal, mitigating the interference from outside. However, it has been clearly shown in [31][32] that even at a distance of 50m between two vehicles, a considerable interference level can still be identified. We have also conducted experiments to test the impact of car body. Our results show that the body can only cause small to negligible differences to packet delivery ratio (i.e., 0.7%). Hence, the inference from neighboring vehicles can indeed degrade the recognition accuracy.

Interference from passengers. When there are multiple passengers together with the driver in the same vehicle, the Doppler frequency shift caused by a passenger might have some impact on the driver side to degrade the performance of feature extraction and quality estimation of the CSI signal. WiMU [33] and CARIN [18] are designed to handle the multi-people scenario. Thus, DriveFusion can be integrated with them if necessary. We focus on the driver only in this paper.

Real-time driver recognition. Existing work on real-time driver recognition [10] can be integrated with DriveFusion for a trade-off between timeliness and recognition accuracy. In addition, CNN-based image feature extraction and learning-based feature fusion, can also be accelerated significantly [34], [35] for real-time recognition.



(a) Devices inside the car (b) Top view illustration

Fig. 7: Testbed car and device placement inside it.

Integration with other sensors. Although designed mainly for fusing camera and CSI signal, DriveFusion provides a general framework that can incorporate other sensor inputs (e.g., wearable sensors), as long as the corresponding feature extraction and signal quality estimation algorithms are similarly designed.

V. Evaluation

In this section, we conduct the evaluation of DriveFusion. We first introduce the experiment setup, and then test DriveFusion against different baselines.

A. Experiment Setup

Testbed Setup and System Implementation: The testbed setup is shown in Figure 7. The testbed of DriveFusion consists of one off-the-shelf WiFi router as the WiFi transmitter, one laptop (the WiFi receiver) equipped with wireless Network Interface Cards (NIC), and two smartphones working as cameras. For the WiFi part, we use TP-Link Archer A7 as the WiFi transmitter, and a ThinkPad T400 with Intel 5300 wireless NICs as the receiver. Wireless packets are generated using the ping command at a transmission rate of 800pkt/s, and CSI samples are collected with the modified network driver on a packet basis [36]. The transmitter is placed in the middle of the dashbroad and the receiver lays between the driving wheel and the panel. For the camera part, we attach one camera on the car ceiling when we test the driver's hand movements (camera location #2). Because this location does not recognize driver's head movements well, we install a second camera on the driver-side sun visor when testing head movements for better recognition results (camera location #1). In a real product, all these devices can be embedded into the car body.

We implement DriveFusion using Keras, a widely-used deep learning framework and train DriveFusion on a GPU RTX 2080Ti with 11GB Memory. The CNN used for image feature extraction is implemented using the model available in Keras, and trained with the image sequence inputs only. The size of the input features from the image and CSI signal is set to 512, and the batch size is set to 20 in the training process. The network is trained using the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.001. Other parameters for training are the default settings.

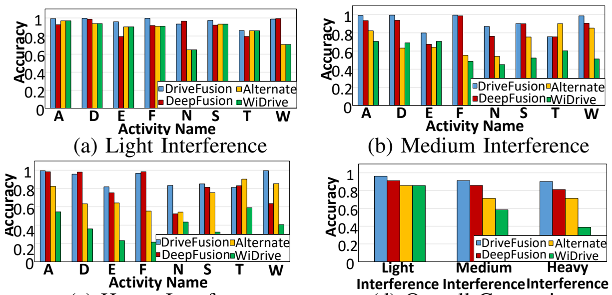
Data Collection and Preparation: We choose eight different typical driver activities as our pre-defined activity set including approaching the driving wheel(**A**), pressing the door button (**D**), eating/drinking (**E**), fetching (**F**), head turning (**H**), nodding off (**N**), swiping hands (**S**), and withdrawing from the driving wheel (**W**). Those investigated activities are chosen based on the safety considerations (nodding off), human-vehicle interactions (swiping hands), and takeover preparations (approaching the driving wheel). All the activities are performed by seven different users in the experiment vehicle, and each driver performs more than 70 times per activity type[1] based on their own understandings of the activity, so there are large variances for stress testing. Though all traces are collected in a parking lot, previous studies have shown that the impact of the vehicle speed is small compared to the driver activities inside the vehicle [10][8].

Baselines: In the experiment, we choose the following activity recognition schemes as the baselines:

- **WiDrive**: WiDrive is a state-of-the-art WiFi-based driver activity recognition scheme [10], which also leverages the time-frequency analysis to extract the feature vectors from CSI signals. However, WiDrive is not designed to handle interference from other WiFi systems, so it may perform poorly when the interference becomes heavy.
- **Alternate**: This baseline is a typical simple combination strategy that alternates between using only camera and using only CSI signal, depending on which one provides a better recognition accuracy. Since it does not consider simultaneously use both the two signal inputs, its performance is expected to be worse than that of DriveFusion. This baseline decides which sensor to used based on the prior knowledge in the experiment when using CSI signal or camera input alone. In reality, it is difficult to decide when to switch from one input to the other without the prior knowledge, so Alternate only serves as an upper bound for similar solutions.
- **DeepFusion**: DeepFusion is a state-of-the-art sensor fusion scheme designed based on deep learning [23]. It features sensor combination and sensor correlation estimation to calculate the weight for each input feature vectors, and a RNN-based activity recognition scheme. However, DeepFusion does not consider time-varying signal qualities, nor does it consider filtering out misleading information from the sensor input when its signal quality is too poor to be used. In contrast, DriveFusion explicitly considers online estimated signal qualities in signal filtering and feature combination. Thus, DriveFusion should outperform DeepFusion when the wireless interference is heavy or the camera image quality is poor.

B. Recognition Accuracy Under Different Interference Levels

Here we compare DriveFusion with the baselines in terms of recognition accuracy under different wireless interference levels. In order to generate different interference levels with just one pair of interfering devices, we adjust the CSI sampling frequency on the receiver side based on the desired interference level with random variances added to our collected CSI traces. In this experiment, we mainly choose three testing scenarios with different CSI sampling frequencies to create the light, medium, and heavy interference levels on the road: The light level corresponding to a CSI sampling frequency of 150Hz with a coefficient of variance of 0.4, which is the same as four interference vehicles in the communication range; The medium level corresponding to a CSI sampling frequency of 100Hz with a coefficient of variance of 1, which represents seven interference vehicles in the communication range; The heavy interference level corresponds to a CSI sampling frequency of 50Hz with a coefficient of variance of 1.4, resembling ten interference vehicles in the range.



(a) Light Interference (b) Medium Interference
(c) Heavy Interference (d) Overall Comparison

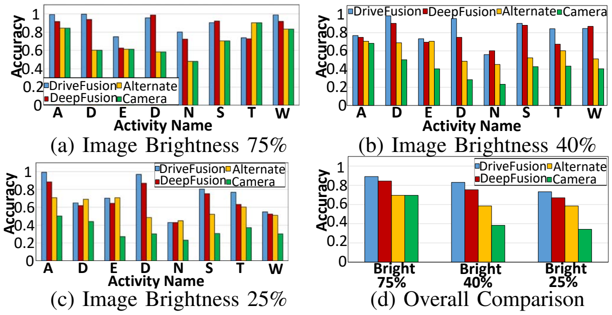
Fig. 8: The performance of DriveFusion and the baselines for the three test cases. DriveFusion achieves a 91.3% recognition accuracy and outperforms the other baselines by 13.4%.

Figure 8 shows the recognition accuracy of DriveFusion and the baselines. DriveFusion outperforms the baseline *DeepFusion* by 6.2% on average, and beats *Alternate* and *WiDrive* by 16.4% and 31.6%, respectively. DriveFusion performs better than *DeepFusion* because DriveFusion is designed specifically for the scenario of fast-changing CSI signal quality due to mobility. As discussed before, DriveFusion dynamically assigns weights to the CSI and image features based on signal quality estimated online. When the wireless interference is so heavy that the CSI signal can hardly provide any useful information, DriveFusion assigns a sufficiently small weight to the CSI feature vector, such that the CSI input does not have a large impact on the recognition performance. Compared with the baseline *Alternate*, the main advantages of DriveFusion lies in its ability to combine the features from both the camera input and CSI signal. As shown in Figure 8(b), even if there is relatively heavy interference where relying on the CSI signal alone can only provide a recognition accuracy of 38.7%, it is still beneficial to combine the CSI input with the camera input to increase the overall recognition performance to 90.2%. Unfortunately, *Alternate* totally discards the CSI signal, so it achieves a recognition accuracy of only 71.3%. At last, compared with *WiDrive*, even under light interference, DriveFusion can still outperform *WiDrive* by 10.4%. This result clearly demonstrates that it is important to 1) fuse CSI with camera for better recognition accuracy, and 2) dynamically assign weight/importance to the two inputs.

C. Different Image Qualities

We now test DriveFusion under different image qualities. Because brightness is a major quality factor, we test images of the same scene under different brightness levels in the same day. Our results with other factors (e.g., fuzziness) are similar and omitted here due to the page limitation. *Bright-75%* represents images with a brightness level that is 75% of the highest brightness; *Bright-40%* has a brightness level of 40%, while *Bright-25%* has only 25%. For the baselines, we exclude *WiDrive* in this experiment because its performance is not influenced by image quality. Instead, we include a new baseline *Camera*, which extracts the features from the image sequence with the pre-processing part in DriveFusion, and

[1]The dataset used in this paper can be downloaded online at: https://www.dropbox.com/s/dfq48jvfmiimnnq/camera_trace_arm.rar?dl=0.



(a) Image Brightness 75% (b) Image Brightness 40%

(c) Image Brightness 25% (d) Overall Comparison

Fig. 9: The performance of DriveFusion and the baselines for different image qualities with medium wireless interference. DriveFusion can still achieve 81.4% accuracy on average.

directly uses the feature sequence to recognize the activity.

Figure 9 shows the recognition accuracy of DriveFusion and the baselines under different image qualities when medium wireless interference is deployed. From Figure 9(a), we can see that DriveFusion has an improvement of 19.6% and 34.7% in terms of recognition accuracy over the baselines *Alternate* and *Camera*, respectively. This result indicates the advantages of using camera together with the WiFi devices over relying only on a single input. However, the accuracy improvement of DriveFusion to *DeepFusion* is only 2.1%, because the image quality with *75%* brightness does not highlight their differences in terms of signal filtering. When the brightness level becomes *40%* and *20%*, respectively, the difference between DriveFusion and *DeepFusion* becomes greater: DriveFusion outperforms *DeepFusion* by 6.3% and 7.6%, respectively. The main reason for the improvement is that DriveFusion features a signal filtering mechanism and is able to prevent any poor image features from feature combination, thus it can avoid degrading the overall recognition performance in the dark and completely dark scenarios. In those scenarios with bad image qualities, the performance of DriveFusion is similar to that of *Alternate*, which relies only on the CSI signal for activity recognition. In summary, as shown in Figure 9(d), DriveFusion outperforms *DeepFusion* by 6.2% due to the signal filtering mechanism, and also beats the *Camera* by 34.4% due to the usage of the WiFi recognition system and feature fusion.

VI. Conclusion

Existing solutions for driver activity recognition rely on either camera or WiFi signal. In this paper, we have proposed DriveFusion, a driver activity recognition system that leverages information in both CSI signals and camera images to perform driver recognition based on a deep learning structure. DriveFusion features an online CSI signal quality estimation methodology, which is built on a regression model with regard to the CSI sampling frequency and variation. We also propose a signal filter that can dynamically determine the best threshold to filter out these input signals with too low signal qualities. DriveFusion also includes a weighted averaging algorithm to combine camera inputs and CSI inputs, where the weights are dynamically calculated with the self-attention mechanism and based on the online estimation to handle fast-changing signal qualities. DriveFusion is evaluated in real cars with commercial off-the-shelf WiFi devices and cameras. Our results show that the average accuracy of DriveFusion is 91.3% when there are different levels of wireless interference, which outperforms the state-of-the-art baselines by 13.4% on average.

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